

Spectral Marker of Feature Interaction in Tian’s Li₂ Grokking Framework

(Technical summary)

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1. Problem

Tian [2025] proposes the Li₂ framework: grokking in 2-layer networks $\hat{Y} = \sigma(XW)V$ ($\sigma(x) = x^2$, MSE loss with sample-mean projector $P_1^\perp$) passes through three stages—Lazy, Independent, and Interactive feature learning—each marked by a different structure of the backpropagated gradient

$$G_F = -\frac{\partial J}{\partial F} = P_1^\perp(Y - FV)V^\top, \quad J = \frac{1}{2} \left\| P_1^\perp(Y - FV) \right\|_F^2. \quad (1)$$

The framework is mathematically clean. *When* transitions between stages happen during training is left explicitly uncharacterized.

2. Result

The rolling-window eigengap σ_2/σ_3 of the parameter-update Gram ΔW is a clean Stage III lock-in detector on Tian’s exact modular addition setup ($M = 71$, $K = 2048$, $n = 2016$, Adam at $\text{lr} = 10^{-3}$, weight decay η). At each epoch we form

$$\Delta := [\Delta W_{t-W+1}, \dots, \Delta W_t] \in \mathbb{R}^{P \times W}, \quad \sigma_k(t) := \lambda_k(\Delta^\top \Delta), \quad (2)$$

with $W = 20$ (a $W \times W$ eigenproblem; $P = 2MK \approx 2.9 \times 10^5$). A slope detector

$$s(t) := \frac{1}{25} [\log(\sigma_2/\sigma_3)(t) - \log(\sigma_2/\sigma_3)(t - 25)], \quad \text{“fire”} := \min\{t \geq 100 : s(t) > 0.04\} \quad (3)$$

gives:

- Fires at **median epoch 174 (IQR = [173, 174]) in 15/15 grok seeds** ($\eta = 2 \times 10^{-4}$) and **0/15 control seeds** ($\eta = 0$).
- Late-stage magnitude separation: $\sigma_2/\sigma_3 \approx 300$ in grok vs ≈ 1.31 in control over epochs 200–400. Cross-condition ratio $229\times$.
- Mechanism: between epochs 150 and 200, σ_2 stabilizes near 5×10^{-3} while σ_3 collapses two orders of magnitude. The rolling spectrum becomes effectively rank-2.

Theorem 6 saturation coincides with the slope-fire moment. Tian [2025] Theorem 6 asserts that for $B := (\tilde{F}^\top \tilde{F} + \eta I)^{-1}$ on similar feature pairs,

$$\text{sgn}(B_{j\ell}) = -\text{sgn}\left(\tilde{f}_j^\top P_{\eta, -j\ell} \tilde{f}_\ell\right), \quad P_{\eta, -j\ell} := I - \tilde{F}_{-j\ell} \left(\tilde{F}_{-j\ell}^\top \tilde{F}_{-j\ell} + \eta I \right)^{-1} \tilde{F}_{-j\ell}^\top. \quad (4)$$

Approximating $P_{\eta, -j\ell}$ by the full projector P_η (relative error $O(2/K)$ at $K = 2048$) and computing B via Woodbury,

$$B = \frac{1}{\eta} I - \frac{1}{\eta^2} \tilde{F}^\top \left(\tilde{F} \tilde{F}^\top + \eta I \right)^{-1} \tilde{F} \quad (\text{reduces } K \times K \text{ inverse to } n \times n), \quad (5)$$

the empirical sign-match on the top-200 most-similar feature pairs at five deterministic-replay checkpoints is:

epoch	50	100	175 (lock-in)	250	300
sign-match	0.83	0.91	0.975	0.975	0.995

The sign-rule saturates (≥ 0.95) precisely at epoch 175—the same epoch the σ_2/σ_3 slope detector fires.

3. Interpretation

In Stage III, similar features repel through the negative off-diagonal entries of B . The rolling ΔW spectrum becomes rank-2 because two persistent feature directions consolidate while the redundant ones are pushed out. σ_2/σ_3 is therefore not just *a* late-stage signature—it is a **direct observable of Tian’s repulsion mechanism becoming dominant**, with the empirical sign-match providing the mechanistic verification.

4. Negative result

A simpler signal—the level metric on the activation Gram,

$$\rho_{\text{tian}}(t) := \frac{\|P_1^\perp F F^\top - (aI + b \mathbf{1}\mathbf{1}^\top)\|_F}{\|F F^\top\|_F}, \quad (6)$$

where a, b are empirical diagonal/off-diagonal averages—fires ~ 80 epochs before test accuracy at the $(M = 71, \eta = 2 \times 10^{-4})$ point. An η **sweep** ($\eta \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}\}$, five seeds each, 600 epochs) preserves this picture: lead is positive in 20/20 grokking runs (range 79–154 epochs); $\eta = 10^{-5}$ does not grok and ρ_{tian} barely touches the 0.075 threshold.

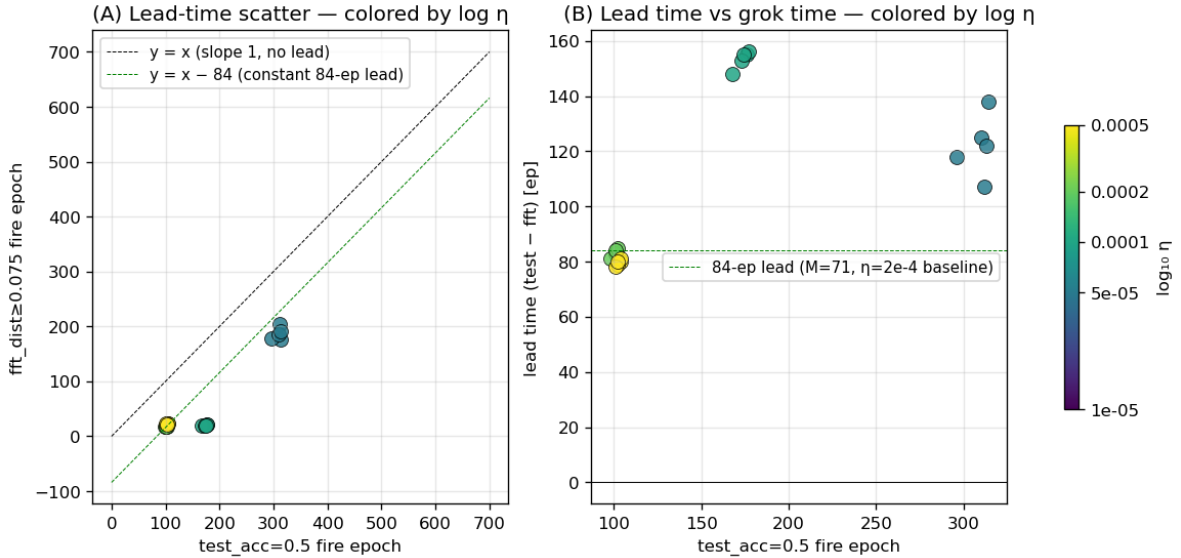


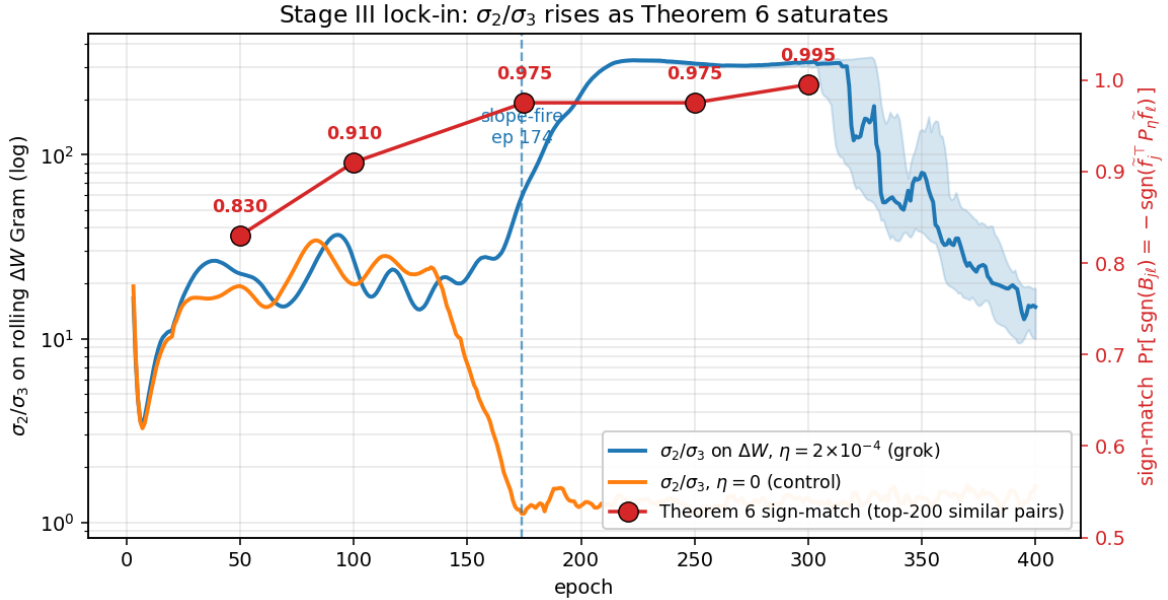
Figure 1: η sweep ($n = 25$). Fft fire epoch versus $t_{\text{test}=0.5}$ (left) and lead time versus $t_{\text{test}=0.5}$ (right), colored by $\log_{10} \eta$. Lead is positive in 20/20 grokking runs. Linear regression: $t_{\text{fft}} = 0.80 t_{\text{test}} - 75$ ($R^2 = 0.87$).

This survives an η -sweep but *not* an $M \times p$ scaling sweep. On a 60-run sweep ($M \in \{41, 71, 127\}$, $p \in \{0.1, 0.2, 0.3, 0.5\}$, five seeds, $\eta = 2 \times 10^{-4}$), $\rho_{\text{tian}} \geq 0.075$ fires in **60/60 runs**

at near-constant epoch (~ 11 – 25), including all cells that fail to grok within the training budget. Across this sweep, fire epoch is decoupled from grokking outcome.

Reconciliation. The η axis is special: η controls whether feature dynamics happen at all. At $\eta = 0$, the gradient signal collapses and ρ_{tian} never crosses threshold. At any $\eta > 0$ large enough to drive feature dynamics, ρ_{tian} crosses—at near-constant epoch regardless of whether (M, p) are sufficient for those dynamics to converge to a generalizable solution. The honest reading: ρ_{tian} is a Stage I→II *initiation* detector—necessary but not sufficient for grokking. Only the lock-in detector σ_2/σ_3 survives the full stress test.

5. Figure



Blue: median σ_2/σ_3 on rolling ΔW Gram across 15 grok seeds ($\eta = 2 \times 10^{-4}$), with IQR shading and the slope-fire epoch (174) marked. **Orange:** median across 15 control seeds ($\eta = 0$). **Red:** Theorem 6 sign-match $\Pr[\text{sgn}(B_{j\ell}) = -\text{sgn}(\tilde{f}_j^\top P_\eta \tilde{f}_\ell)]$ on top-200 most-similar feature pairs at five deterministic-replay checkpoints. The σ_2/σ_3 rise to its post-grokking plateau coincides exactly with Theorem 6 sign-match crossing 0.95.

References

Yuandong Tian. Provable scaling laws of feature emergence from learning dynamics of grokking. *arXiv preprint arXiv:2509.21519*, 2025. URL <https://arxiv.org/abs/2509.21519>.